

Glove gap: what work gloves do to the 3D hand-pose front end of human-video → robot learning

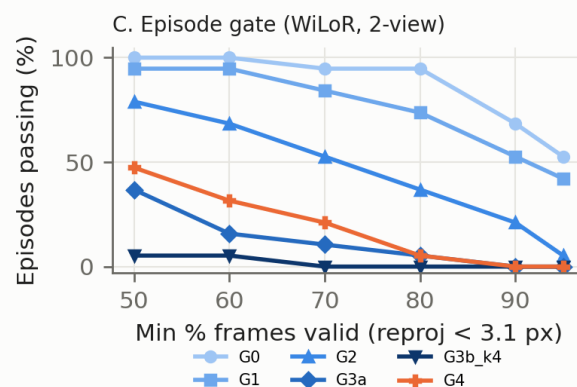
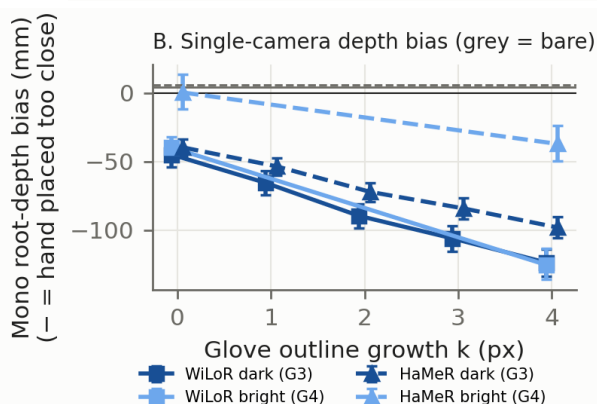
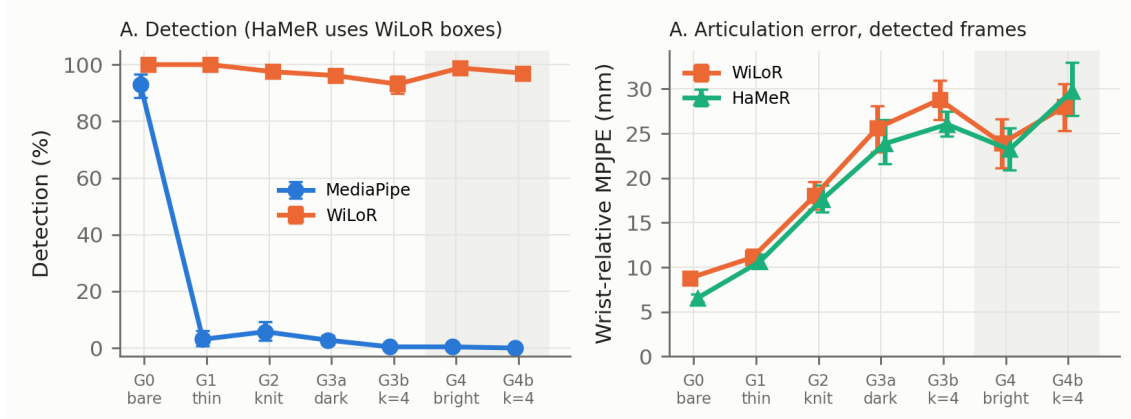
How much does 3D hand-pose estimation degrade when the operator wears work gloves, and which capture-spec choices recover it?

Setup

Data: DexYCB subject-01, 19 held-out S0-val grasp sequences (+1 in appendix), GT 3D joints; a 2-view rig (1.03 m baseline, 70°) plus an 8-camera bound. **Gloves** (synthetic, on refined hand masks): G0 bare, G1 thin bright nitrile, G2 grey knit, G3a dark insulated, G3b = G3a + outline growth $k = 1-4$ px, G4 = bright insulated ($k = 0, 4$). **Models:** MediaPipe Hands, WiLoR, HaMeR (on WiLoR boxes). **Real-glove check:** Roboflow photos, 83 images. CIs: bootstrap over sequences (95%); Wilson for recall.

Findings

- a Real gloves break detection; synthetic gloves underestimate it.** On real worn-glove photos WiLoR's detector finds 52–65% of gloves vs 100% of bare hands (–35 to –48 pp, depending on how strictly boxes must match: glove boxes include the cuff); MediaPipe finds 7–13%. On synthetic gloves WiLoR loses only 0–4 pp (MediaPipe 87–90 pp), so synthetic detection numbers for WiLoR are optimistic.
- b Gloves bias single-camera depth.** Monocular WiLoR places the hand 45 mm too close with a dark insulated glove and about 20 mm more per px of outline growth (–124 mm at $k = 4$). Bright colour does not remove it for WiLoR (G4: –41 mm). Take hand position from depth or ≥ 2 views, never from one RGB camera.
- c Multi-view fixes position, not articulation.** 2-view triangulation removes 72–88% of the glove-induced position gap (8 cameras: 83–93%) but only 3–49% of the articulation (wrist-relative) gap, for both WiLoR and HaMeR. Articulation is set by the glove: thin nitrile instead of the work glove removes 73–92% of it, while bright colour on an insulated glove does not help (G4 vs G3a: –1.7 mm [–3.6, +0.1]).
- d A reprojection gate is a batch-level glove alarm, not an episode ranker.** With frames valid at 2-view reprojection < 3.1 px and episodes passing at $\geq 80\%$ valid frames: G0 95%, G1 74%, G2 37%, G3a 5%, G4 5% pass. Within a glove type, the valid fraction does not predict error ($|p| \leq 0.36$, $p > 0.1$).



Capture spec	Number	Evidence (this PoC)
Pose model	VIT/MANO class (WiLoR, HaMeR); not MediaPipe	MediaPipe detection 93% → ≤6% synthetic, 7% real
Glove	Thin, form-fitting nitrile; colour secondary	G1: PA-MPJPE +0.6 mm vs bare; insulated +2.4 mm and wrist-rel ×2.9
Thickness	Minimise: each px of apparent growth = +0.4 mm PA-MPJPE, –20 mm mono depth	G3b sweep k = 0–4
Views	≥2 calibrated views, ~1 m baseline, ~70° (or RGB-D for position)	Dark glove, position error 14.6 mm (2-view) vs 48.8 mm (mono)
Gate	Per batch: $\tau = 3.1$ px, $X = 80\%$; investigate batches below ~75% pass	Pass 95% bare → 37% knit → 5% insulated

Limits. Synthetic gloves are 2D image edits: no 3D thickening, finger gaps close, mask-edge residuals. Possible training leakage: HaMeR (and likely WiLoR) train on DexYCB S0-train; held-out S0-val may have been used for model selection, so relative degradation is the claim, not absolute accuracy. One subject, YCB lab objects, room temperature, not a warehouse. The real-glove check is detection-only, single-frame, $n = 83$ images (31 worn-glove boxes). Grasp-onset F1 is secondary: the object-motion label is noisy (GT-aperture ceiling 0.79).

Code: make all reproduces every number and figure from cached predictions. Licences: DexYCB CC BY-NC 4.0; WiLoR CC-BY-NC-ND (+ MANO, Ultralytics); HaMeR MIT (+ MANO); Roboflow 'Gloves and bare hands detection' CC BY 4.0. Non-commercial research PoC.